La Scuola d'Italia / Dr. Huson / IB Math 12 HL
14 October 2026

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**12.6 Quiz: Complex Numbers**

Answer all five problems. Use the spaces provided for Problems 1 and 2. Answer Problems 3–5 on lined paper.

1. [Maximum mark: 6]

   Let $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$.

   (a) Find $z_1 z_2^*$, giving your answer in the form $a + bi$, where $a, b \in \mathbb{R}$. [2]

   (b) Find $\dfrac{z_1}{z_2}$, giving your answer in the form $a + bi$, where $a, b \in \mathbb{R}$. [2]

   (c) Find the modulus and the argument of $\dfrac{z_1}{z_2}$. [2]

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2. [Maximum mark: 7]

Let $z = -\sqrt{3} + i$ and $w = 3\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$.

(a) Write $z$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [2]

(b) Find $zw$ in the form $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $-\pi < \theta \leq \pi$. [2]

(c) Write $zw$ in the form $a + bi$. [1]

(d) On an Argand diagram, the point representing $z$ is mapped to the point representing $zw$. Describe this transformation geometrically. [2]

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*Answer Problems 3–5 on lined paper.*

3. [Maximum mark: 13]

The equation

$$z^3 + az^2 + bz + 26 = 0,$$

where $a, b \in \mathbb{R}$, has a root $3 + 2i$.

(a) Write down a second root of the equation, giving a reason for your answer. [2]

(b) By considering the product of the roots, find the third root. [3]

(c) Find the value of $a$ and the value of $b$. [4]

The three roots are represented by the points A, B and C on an Argand diagram, where A represents $3 + 2i$ and C represents the real root.

(d) Show that triangle ABC is isosceles. [2]

(e) Find the area of triangle ABC. [2]

4. [Maximum mark: 14]

(a) Use mathematical induction to prove that

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

for all $n \in \mathbb{Z}^+$. [6]

(b) By using de Moivre's theorem with $n = 4$, show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$. [5]

(c) Use de Moivre's theorem to find $\left(1 - \sqrt{3}\,i\right)^{10}$, giving your answer in the form $a + b\sqrt{3}\,i$, where $a, b \in \mathbb{Z}$. [3]

5. [Maximum mark: 13]

Let $w = -4\sqrt{2} + 4\sqrt{2}\,i$.

(a) Write $w$ in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [2]

(b) Solve the equation $z^3 = w$, giving each solution in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [5]

(c) Write the solution that lies in the first quadrant in the form $a + bi$. [1]

(d) Show that the sum of the three solutions is zero. [2]

(e) Find the three values of $w^{\frac{2}{3}}$, giving each in the form $a + bi$, where $a, b \in \mathbb{R}$. [3]